Computer Vision — Practical Laboratory

Lab 1: Image Labeling and Annotation

Using Roboflow

Academic level: Master's — Computer Science / Artificial Intelligence
Duration: 3 hours
Platform: Roboflow
Lab type: Hands-on practical work

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1 Lab Objectives

This laboratory introduces four computer vision annotation tasks using Roboflow. By the end of the lab, students should be able to:

1. Label images for **image classification**.
2. Annotate objects with bounding boxes for **object detection**.
3. Label image regions for **semantic segmentation**.
4. Identify and annotate landmarks for **keypoint detection**.
5. Explain how the required annotations differ across these four tasks.

2 Practical Exercises

2.1 Exercise 1: Image Classification

Objective: Assign an appropriate class label to each image.

Instructions

1. Open Roboflow and create a new project.
2. Select **Single-Label Classification**.
3. Name the project `Lab1_Image_Classification`.
4. Upload at least **15 images** representing three categories: **Cat**, **Dog**, and **Bird**.
5. Assign the appropriate class label to each image.
6. Review and save the labels.

Checkpoint 1 — Image Classification

☐ The classification project has been created.

☐ At least 15 images have been uploaded.

☐ Three classes have been defined.

☐ Every image has been correctly labeled.

Checkpoint question: How does single-label image classification handle an image containing multiple objects?

2.2 Exercise 2: Object Detection

Objective: Locate objects and associate them with class labels using bounding boxes.

Instructions

1. Create a new Roboflow project of type **Object Detection**.
2. Name the project `Lab1_Object_Detection`.
3. Upload at least **10 images** containing multiple objects.
4. Define three classes: **Person**, **Car**, and **Bicycle**.
5. Draw bounding boxes around all visible objects belonging to these classes.
6. Assign the correct label to each bounding box.
7. Review and save the annotations.

Checkpoint 2 — Object Detection

☐ The object detection project has been created.

☐ At least 10 images have been annotated.

☐ All relevant objects have bounding boxes.

☐ Bounding boxes accurately surround their corresponding objects.

☐ Each bounding box has the correct class label.

Checkpoint question: What information does object detection provide that image classification does not?

2.3 Exercise 3: Semantic Segmentation

Objective: Assign semantic labels to image regions (pixels) without differentiating individual object instances.

Instructions

1. Create a new Roboflow **segmentation** project, named `Lab1_Semantic_Segmentation`.
2. Upload at least **5 images** containing several semantic regions.
3. Define three classes: **Road**, **Building**, and **Vegetation**.
4. Use the segmentation annotation tools to mark the image regions belonging to each class.
5. Give regions belonging to the same semantic class the same class label.
6. Review and save the annotations.

Note: Where the platform represents segmentation annotations as separate regions or polygons, the intended semantic result is a class-level pixel mask, not individual instance IDs.

Checkpoint 3 — Semantic Segmentation

☐ The segmentation project has been created.

☐ At least 5 images have been annotated.

☐ Three semantic classes have been defined.

☐ Relevant image regions have been correctly annotated.

☐ Regions belonging to the same class share the same semantic label.

Checkpoint question: If an image contains three cars, can semantic segmentation distinguish each car individually? Explain.

2.4 Exercise 4: Keypoint Detection

Objective: Identify and annotate specific anatomical landmarks on human bodies.

Instructions

1. Create a Roboflow project supporting **Keypoint Detection** (or keypoint annotation).
2. Name the project `Lab1_Keypoint_Detection`.
3. Upload at least **5 images** depicting people in different poses.
4. Define the object class **Person**.
5. Configure seven landmarks: **Head**, **Left shoulder**, **Right shoulder**, **Left elbow**, **Right elbow**, **Left wrist**, and **Right wrist**.
6. Annotate the landmarks on every relevant visible person.
7. Review and save the annotations; handle occluded or invisible landmarks consistently with the tool's visibility options.

Checkpoint 4 — Keypoint Detection

☐ The keypoint detection project has been created.

☐ At least 5 images have been annotated.

☐ All seven keypoints have been defined.

☐ Visible keypoints have been placed accurately.

☐ Each person has an independent set of keypoint annotations.

☐ Invisible keypoints have been handled consistently.

Checkpoint question: Why is keypoint detection more suitable than object detection for estimating human body posture?

3 Final Checkpoint: Comparing Computer Vision Tasks

Scenario: Consider an image containing **three cars and two pedestrians**. Complete Table 1 based on the four tasks. **Questions**

1. Which task assigns a category to an entire image?
2. Which task identifies the locations of individual objects?
3. Which task assigns a semantic category to each pixel?
4. Which task identifies specific anatomical landmarks on a human body?
5. Can two people belonging to the same class have different keypoint configurations?
6. Why is annotation consistency important when preparing a computer vision dataset?

Table 1: Task comparison worksheet

Task	What should be annotated?	Expected output
Image Classification		
Object Detection		
Semantic Segmentation		
Keypoint Detection		

Table 2: Minimum practical deliverables

Deliverable	Minimum requirement
Image Classification	15 labeled images
Object Detection	10 annotated images
Semantic Segmentation	5 annotated images
Keypoint Detection	5 annotated images

4 Lab Deliverables

Submit four Roboflow projects satisfying Table 2. For each exercise, submit the **Roboflow project link**, **screenshots of completed annotations**, and **answers to the checkpoint questions**. Also complete the final checkpoint table and questions.

5 Assessment

Component	Weight
Image Classification	20%
Object Detection	25%
Semantic Segmentation	25%
Keypoint Detection	25%
Final Checkpoint	5%
Total	**100%**

6 References

- Roboflow platform: https://roboflow.com/
- Roboflow documentation: https://docs.roboflow.com/

Scope: This lab covers labeling and annotation only: image classification, object detection, semantic segmentation, and keypoint detection. Model training, preprocessing, augmentation, and instance segmentation are excluded.